

DSA0306-Natural Language Processing

Experiments

1. Artificial Intelligence (AI) is transforming industries across the world. AI is used in healthcare to assist doctors in diagnosis, in banking to detect fraud, and in education to provide personalized learning experiences. Many companies invest heavily in AI research because AI improves efficiency and enables intelligent decision-making. As AI continues to evolve, professionals with AI skills are in high demand.

Write a Python program using the re module to perform the following operations for the word "AI":

- 1. Find the first occurrence of the word "AI" using re.search().**
- 2. Display the starting position of the first occurrence.**
- 3. Display the ending position of the first occurrence.**
- 4. Display the total number of times the word "AI" appears in the passage.**
- 5. Print an appropriate message if the word is not found.**

Program:

```
import re
text="Artificial Intelligence (AI) is transforming industries across the world. AI is used in healthcare to assist doctors in diagnosis, in banking to de
result=re.search(r"\bAI\b",text)
if result:
    print("first occurrence of 'AI' found.")
    print("Starting Position:",result.start())
    print("Ending Position:",result.end())
    count=len(re.findall(r"\bAI\b",text))
    print("total occurrences:",count)
else:
    print("the word 'AI' was not found.")
```

Output:

```
>>>
===== RESTART: C:/Users/hi/AppData/Local/Programs/Python/Python313/ex 1.py =====
first occurrence of 'AI' found.
Starting Position: 25
Ending Position: 27
total occurrences: 6
>>>
```

2. Refer the previous passage and Write a Python program using the re.split() method to perform the following operations:

Split the passage into sentences using appropriate punctuation (., ?, !) as delimiters.

Count and display the total number of sentences.

Split the passage into words by considering one or more whitespace characters as delimiters.

Count and display the total number of words.

Display the list of extracted sentences and words.

Program:

```
import re
text="""Artificial Intelligence (AI) is transforming industries across the world. AI is used in healthcare to assist doctors in diagnosis, in banking to detect fraud, and in education to provide personalized learning experiences. Many companies invest heavily in AI research because AI improves efficiency and enables intelligent decision-making. As AI continues to evolve, professionals with AI skills are in high demand"""
sentences=re.split(r'[.!?]+',text)
sentences=[s.strip() for s in sentences if s.strip()]
print("sentences:")
print(sentences)
print("\n total number of sentences:",len(sentences))
words=re.split(r'\s+',text)
print("\n words:")
print(words)
print("\ntotal no of words:",len(words))
```

Output:

```
===== RESTART: C:/Users/hi/AppData/Local/Programs/Python/Python313/exp 2.py =====
sentences:
['Artificial Intelligence (AI) is transforming industries across the world', 'AI is used in healthcare to assist doctors in diagnosis, in banking to detect fraud, and in education to provide personalized learning experiences', 'Many companies invest heavily in AI research because AI improves efficiency and enables intelligent decision-making', 'As AI continues to evolve, professionals with AI skills are in high demand']

total number of sentences: 4

words:
['Artificial', 'Intelligence', '(AI)', 'is', 'transforming', 'industries', 'across', 'the', 'world.', 'AI', 'is', 'used', 'in', 'healthcare', 'to', 'assist', 'doctors', 'in', 'diagnosis,', 'in', 'banking', 'to', 'detect', 'fraud,', 'and', 'in', 'education', 'to', 'provide', 'personalized', 'learning', 'experiences.', 'Many', 'companies', 'invest', 'heavily', 'in', 'AI', 'research', 'because', 'AI', 'improves', 'efficiency', 'and', 'enables', 'intelligent', 'decision-making.', 'As', 'AI', 'continues', 'to', 'evolve,', 'professionals', 'with', 'AI', 'skills', 'are', 'in', 'high', 'demand.']
```

3. A website requires users to register by providing an Email ID and a Strong Password. Write a Python program using the re module to validate the following inputs.

Validation Criteria:

Email ID

- **Must begin with one or more letters or digits.**
- **May contain ., _, %, +, or - before the @ symbol.**
- **Must contain a valid domain name.**
- **Must end with a valid domain extension such as .com, .org, .edu, or .in.**

Strong Password

- **Must contain at least 8 characters.**
- **Must contain at least one uppercase letter.**
- **Must contain at least one lowercase letter.**
- **Must contain at least one digit.**

- Must contain at least one special character from @, #, \$, %, &, !, or *.
- Must not contain any whitespace characters.

Program

```
exp 3.py - C:/Users/hi/AppData/Local/Programs/Python/Python313/exp 3.py (3.13.2)
File Edit Format Run Options Window Help
import re

# Get input from user
email = input("Enter Email ID: ")
password = input("Enter Password: ")

# Email Regex Pattern
email_pattern = r'^[A-Za-z0-9][A-Za-z0-9._%+-]*@[A-Za-z0-9.-]+\.(com|org|edu|

# Password Regex Pattern
password_pattern = r'^(?=.*[A-Z])(?=.*[a-z])(?=.*\d)(?=.*[@#$$%&!*])[A-Za-z\d@

# Validate Email
if re.fullmatch(email_pattern, email):
    print("Valid Email ID")
else:
    print("Invalid Email ID")

# Validate Password
if re.fullmatch(password_pattern, password):
    print("Strong Password")
else:
    print("Invalid Password")
```

Output:

```
Python 3.13.2 (tags/v3.13.2:4f8bb39, Feb 4 2025, 15:23:48) [MSC v.1942 64 bit (
AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
===== RESTART: C:/Users/hi/AppData/Local/Programs/Python/Python313/exp 3.py =====
Enter Email ID: deekshasree2007@gmail.com
Enter Password: 5237399
Valid Email ID
Invalid Password
>>>
```

5.

Program:

```
exp 5.py - C:/Users/hi/AppData/Local/Programs/Python/Python313/exp 5.py (3.13.2)
File Edit Format Run Options Window Help
import re

# Compile the regular expression only once
email_pattern = re.compile(r'^[A-Za-z0-9._%+-]*@[A-Za-z0-9.-]+\.[A-Za-z]{2,}$')

# Number of email IDs
n = int(input("Enter the number of email IDs: "))

# Validate each email
for i in range(n):
    email = input(f"Enter email {i+1}: ")

    if email_pattern.fullmatch(email):
        print(email, "-> Valid Email")
    else:
        print(email, "-> Invalid Email")
```

Output:

```
Edit Shell Debug Options Window Help
Python 3.13.2 (tags/v3.13.2:4f8bb39, Feb  4 2025, 15:23:48) [MSC v.1942 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.

==== RESTART: C:/Users/hi/AppData/Local/Programs/Python/Python313/exp 5.py ====
Enter the number of email IDs: 2
Enter email 1: regvm,,,
regvm,,, -> Invalid Email
Enter email 2: ytuhkm
ytuhkm -> Invalid Email
```